

Memo Test 4.

$$(1.1) \quad f'(x) = ex^{e-1} + e^x$$

$$(1.2) \quad f'(x) = \tan x + x \sec^2 x$$

$$(1.3) \quad f'(x) = \frac{(\cos x + 3)(\cos x) - (\sin x + 1)(-\sin x)}{(\cos x + 3)^2}$$

$$(1.4) \quad f'(x) = 7(x^3 \sin x)^6 \times [3x^2 \sin x + x^3 \cos x]$$

$$(1.5) \quad f'(x) = -\sin x^2 \times 2x - \frac{1}{x^2}$$

$$(1.6) \quad f(x) = \sin(\cos(\tan x))$$

$$\begin{aligned} \Rightarrow f'(x) &= \cos(\cos(\tan x)) \times \frac{d}{dx}(\cos(\tan x)) \\ &= \cos(\cos(\tan x)) \times -\sin(\tan x) \times \frac{d}{dx}(\tan x) \\ &= \cos(\cos(\tan x)) \times (-\sin \tan x) \times \sec^2 x \end{aligned}$$

$$(1.7) \quad f'(x) = \cosh\left(\frac{x}{3}\right) \times \frac{1}{3}$$

$$(1.8) \quad f'(x) = 4 \sec^3(3x+2) \times (\tan(3x+2) \sec(3x+2)) \times 3$$

$$(2.1) \quad y = (\tan^{-1} x)^2$$

$$\Rightarrow \ln y = \ln(\tan^{-1} x)^2 \\ = 2 \ln(\tan^{-1} x)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{d}{dx} [2 \ln(\tan^{-1} x)] \\ = \ln(\tan^{-1} x) + 2 \frac{1}{\tan^{-1} x} \times \frac{d}{dx}(\tan^{-1} x) \\ = \ln(\tan^{-1} x) + \frac{2}{\tan^{-1} x} \times \frac{1}{1+x^2}$$

$$\Rightarrow \frac{dy}{dx} = (\tan^{-1} x)^2 \left[\ln(\tan^{-1} x) + \frac{2}{(1+x^2) \tan^{-1} x} \right]$$

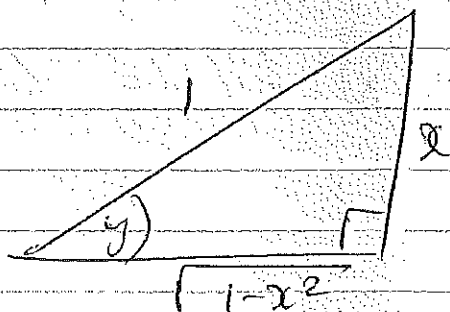
$$(2.2) \quad y = \arcsin x$$

$$\Rightarrow \sin y = x, \quad y \in [-\pi/2, \pi/2]$$

$$\frac{d}{dx}(\sin y) = \frac{d}{dx} x$$

$$\Rightarrow \cos y \times \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\cos y} \\ = \frac{1}{\sqrt{1-x^2}}$$



$$\cos y = \sqrt{1-x^2} \\ y \in [-\pi/2, \pi/2]$$

Question 3

$$f(x) = \sqrt{3x^2 - 2}$$

$$\begin{aligned} f'(x) &= \frac{1}{2}(3x^2 - 2)^{-1/2} \times 6x \\ &= 3x(3x^2 - 2)^{-1/2} \end{aligned}$$

$$f'(1) = 3(3 - 2)^{-1/2} = 3$$

When $x = 1$

$$f(1) = \sqrt{3 - 2} = 1$$

⇒ Equation of tangent:

$$(y - 1) = m(x - 1)$$

$$= 3(x - 1) = 3x - 3$$

$$\Rightarrow \underline{y = 3x - 2}$$